Measure of Central Tendency

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Probability to Statistics (Descriptive vs. Inferential)

👉 Probability: A field of mathematics where we predicts the likelihood of future events. Probability is a number between 0 and 1, where a higher number means a more likely event.

👉 Statistics: A field of mathematics that analyzes the frequency of past events, which involves collecting, organizing, displaying, analyzing, interpreting, and presenting data.
There are two main types of statistics: descriptive statistics & inferential statistics

👉 Descriptive Statistics **: Describes various properties to measure & represent sample and population data.

👉 Statistical Inference: Statistical inference is based on probability and involves using a statistic to draw a conclusion about a parameter.

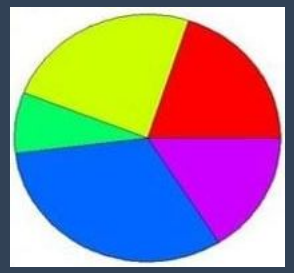
Probability to Statistics (Descriptive vs. Inferential)

- **Stock Price Outcome:**

- **Stock Price Descriptive Statistics:**

- **Stock Price Inferential Statistics:**

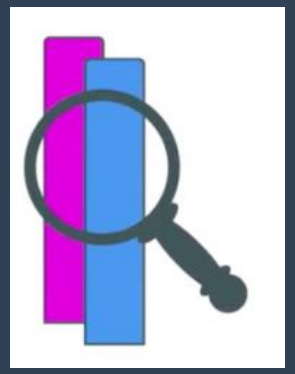
Statistics



- A statistic is a value that has been produced from a data collection, such as a summary measure, an estimate or projection.

- Statistics is the science concerned with developing and studying methods for collecting, analyzing, interpreting and presenting empirical data.

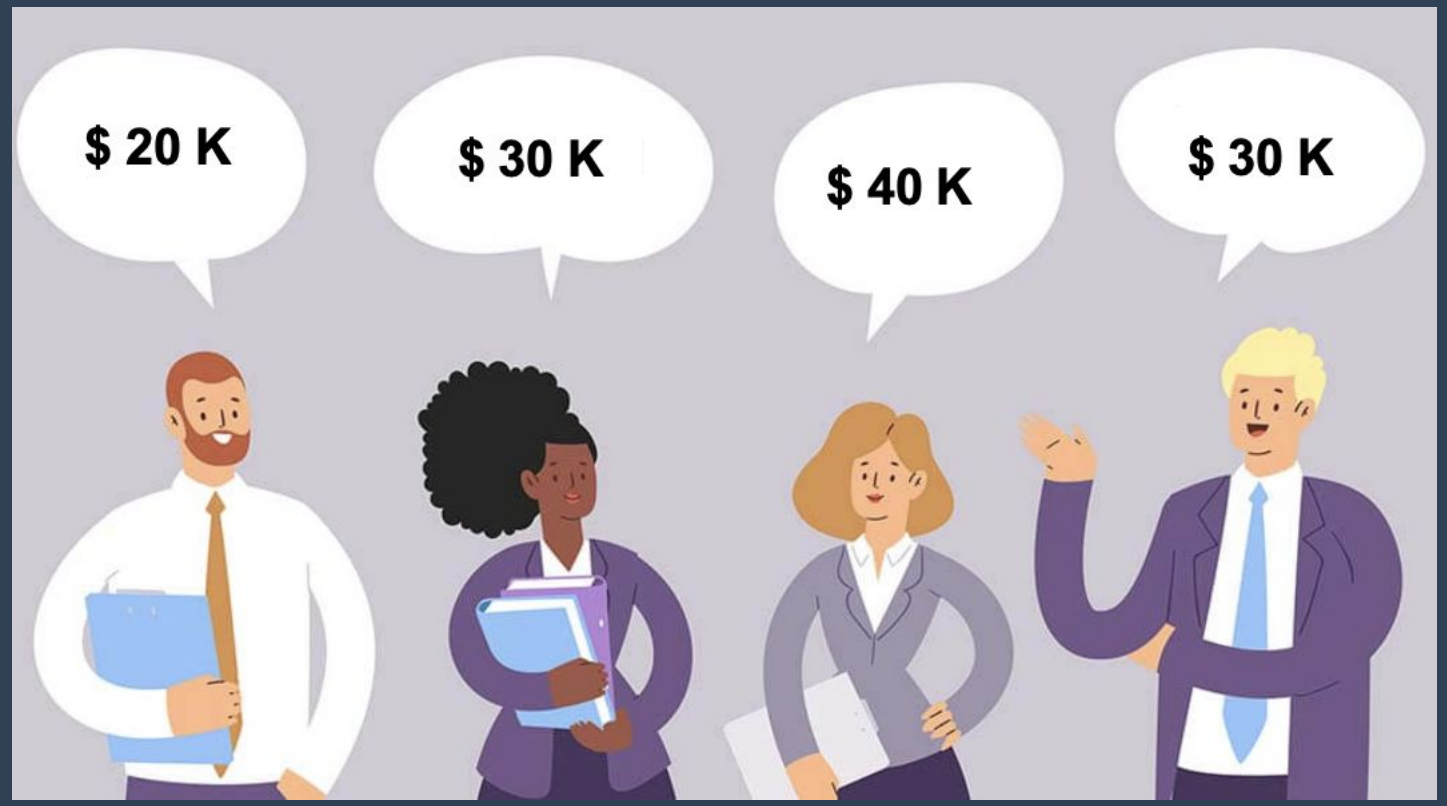
- Statistical information is set of metrics that has been organized to serve a useful purpose.



Statistics

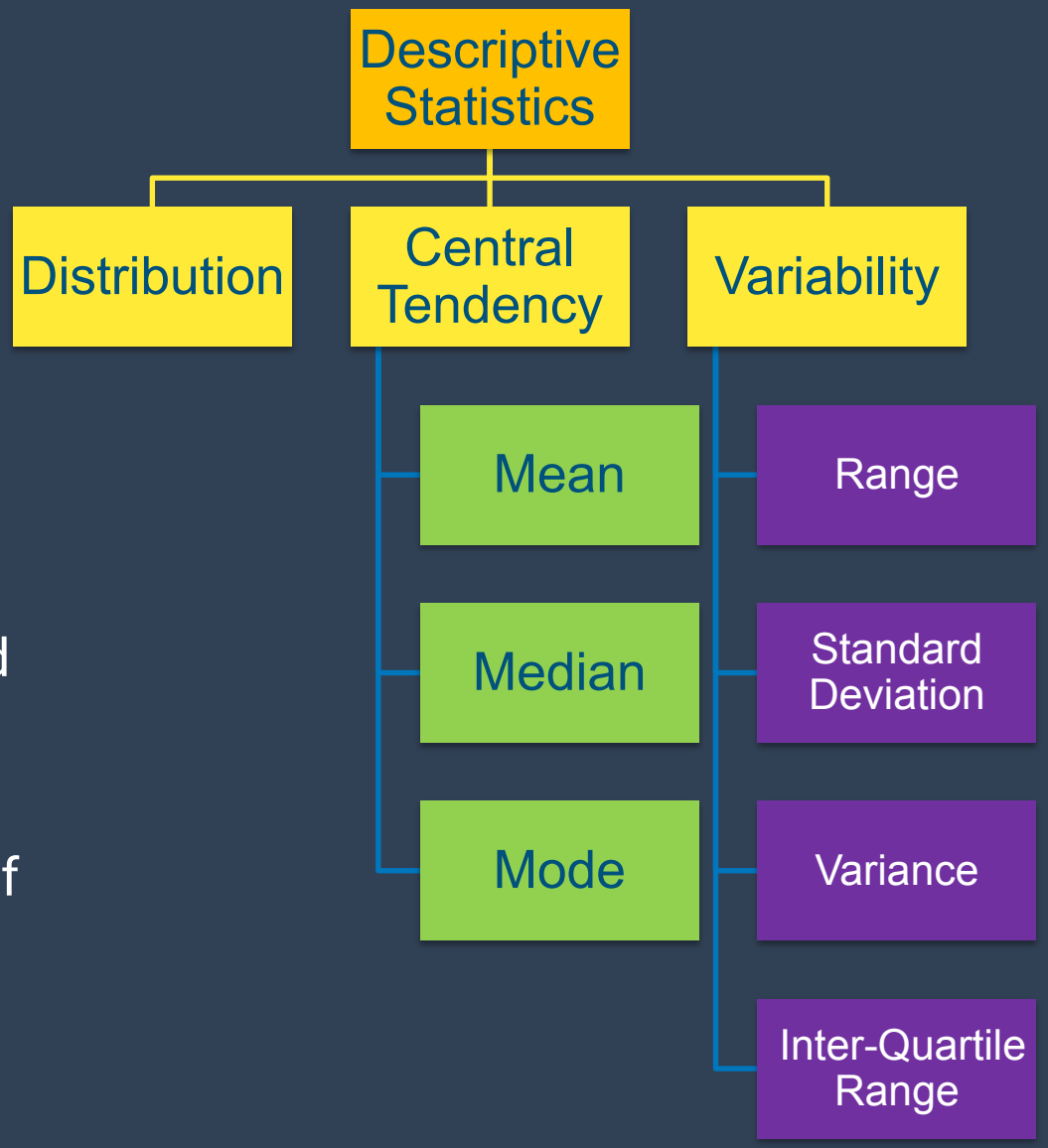
Statistics

- How to summarize the income of your friends.



Descriptive Statistics

- Descriptive statistics summarizes or describes the characteristics of a data set.
- Descriptive statistics consists of three basic categories of measures: measures of central tendency **, measures of variability (or spread), and frequency distribution.
- Measures of central tendency describe the center of the data set (***mean, median, mode***).



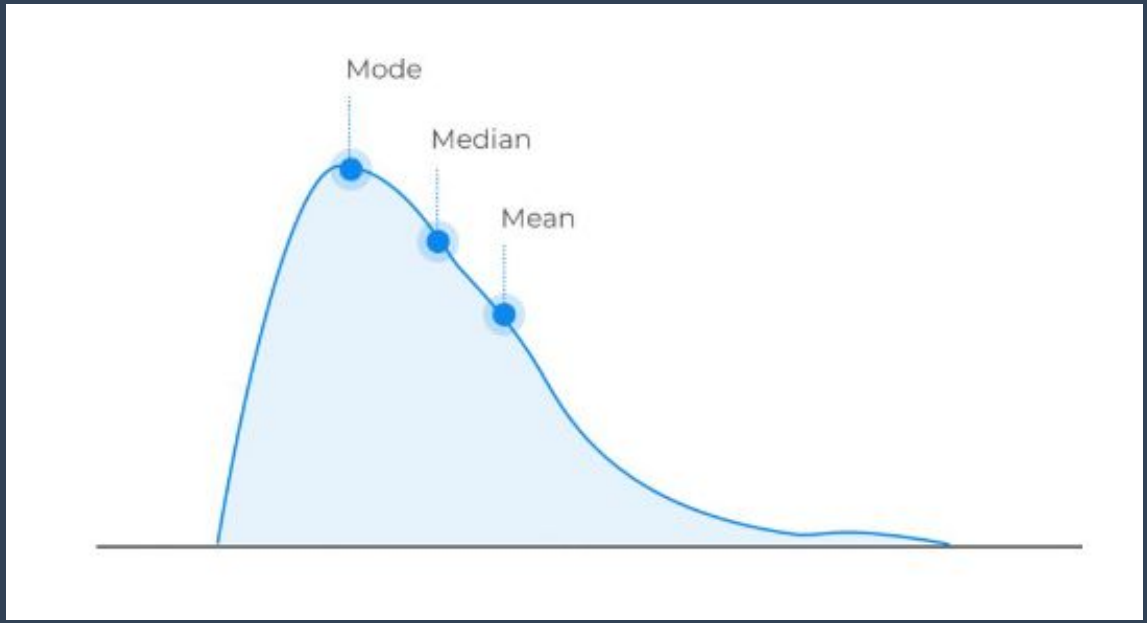
Descriptive Statistics

- How to summarize the income of your friends in the dinner party ?

Central Tendency

- Measures of central tendency help describe and understand the features of a specific data set by giving short summaries about the sample and measures of the data.

- Measures of central tendency focus on the average or middle values of data sets, for example describe the center position of a distribution for a data set.



Mean/Median/Mode

Mean

Commonly referred to as (arithmetic) average.
Sum of observations divided by the number of observations.

Median

Middle value when data are arranged in numeric order.
Average of the two middle numbers when number of observations is even.

Mode

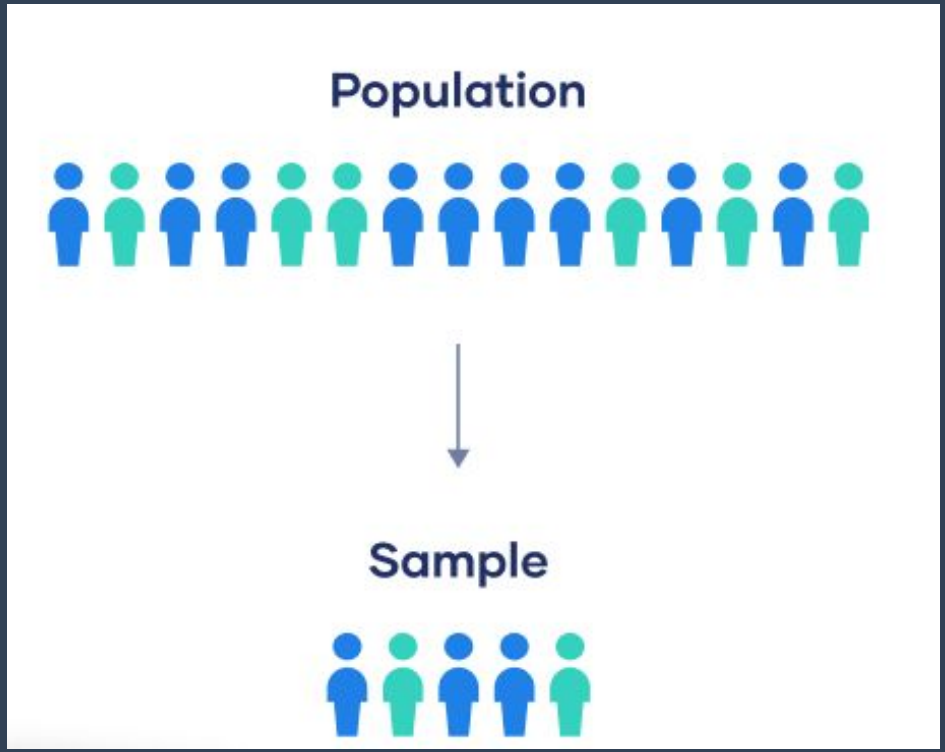
The value that occurs most frequently in a set of data

Population vs Sample

👉 A ***population*** is the entire group that you want to draw conclusions about.
E.g. Undergraduate students in the USA

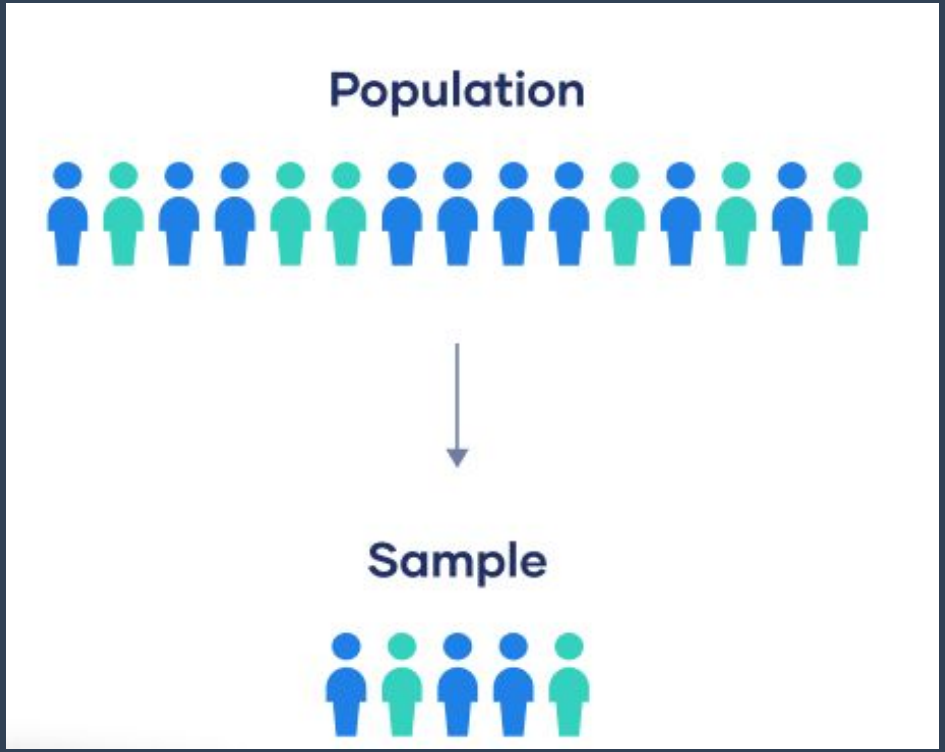
👉 A ***sample*** is the specific group that you will collect data from. The size of the sample is always less than the total size of the population.
E.g. 300 undergraduate students from three US universities who volunteer for your AI research study

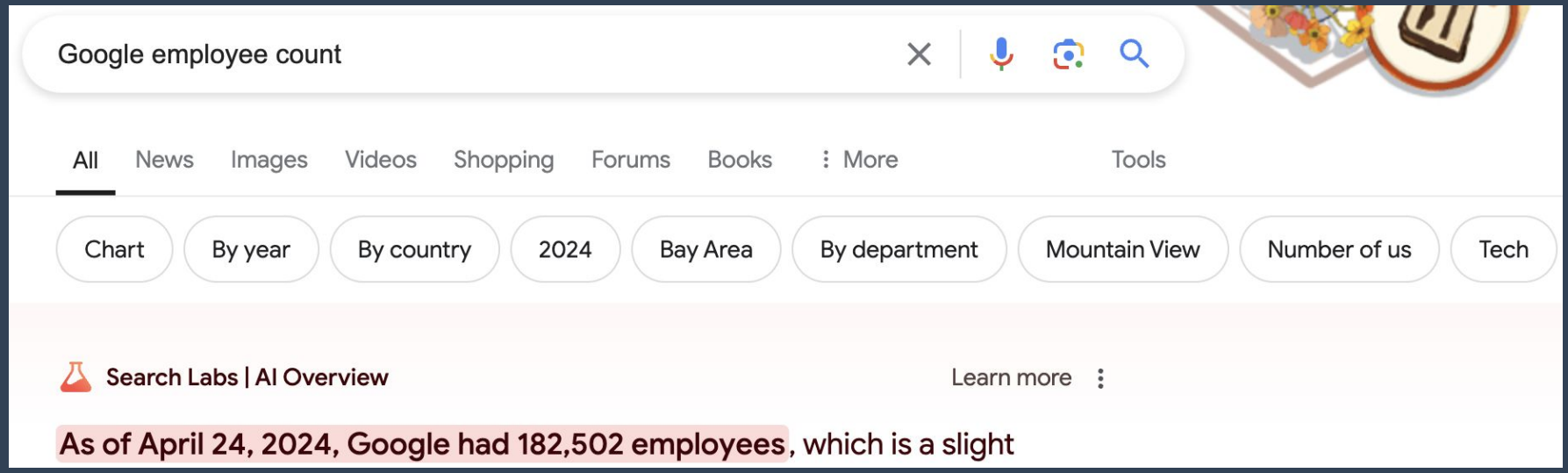
*Why we need to **Sampling**?*



Population vs Sample

Why we need to ***Sampling***?





Mean

- The ***population mean*** and the **sample mean**, both are ***arithmetic means***.

- The ***arithmetic mean*** for any data set is unique and is computed using all the data values. Among all the measures of central tendency, it is the only measure for which the sum of the deviations from the mean is zero.

$$\text{Population mean} = \frac{\sum X_i}{N}$$

$$\text{Sample mean} = \frac{\sum X_i}{n}$$

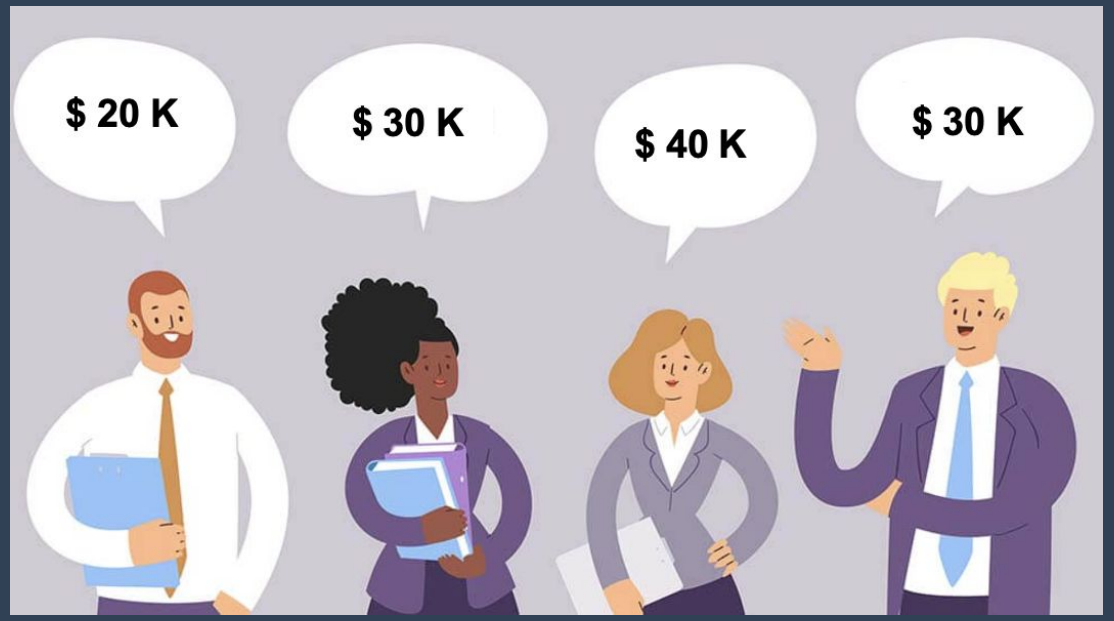
Mean

Let's find out **Mean Salary**

The ***mean (average)*** of a data set is found by adding all numbers in the data set and then dividing by the number of values in the set.

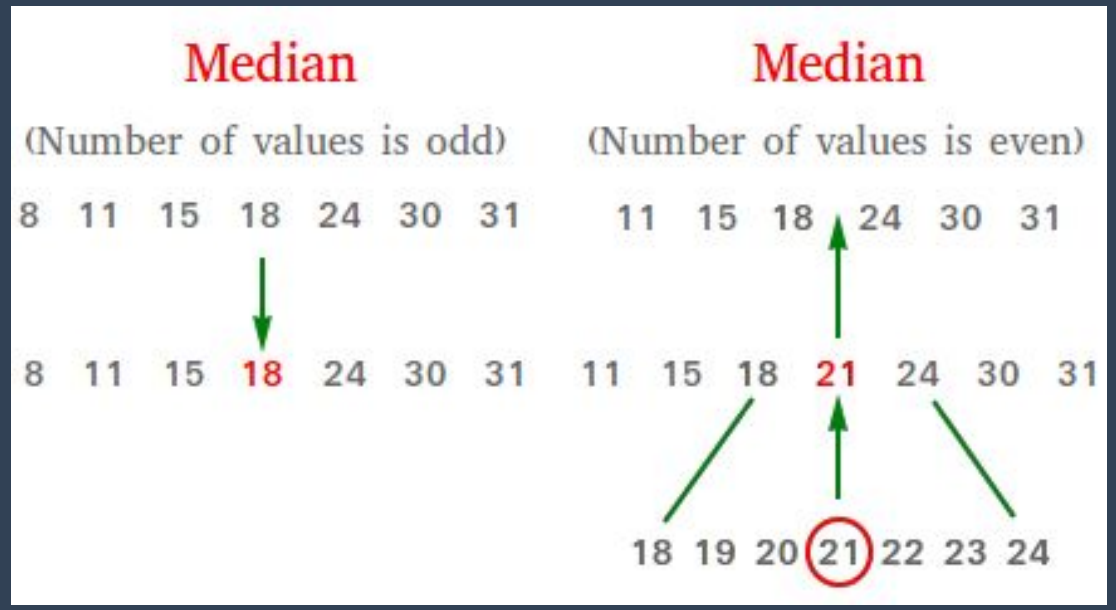
$$Mean = \frac{\$20K + \$30K + \$40K + \$30K}{4}$$

Mean = $30K



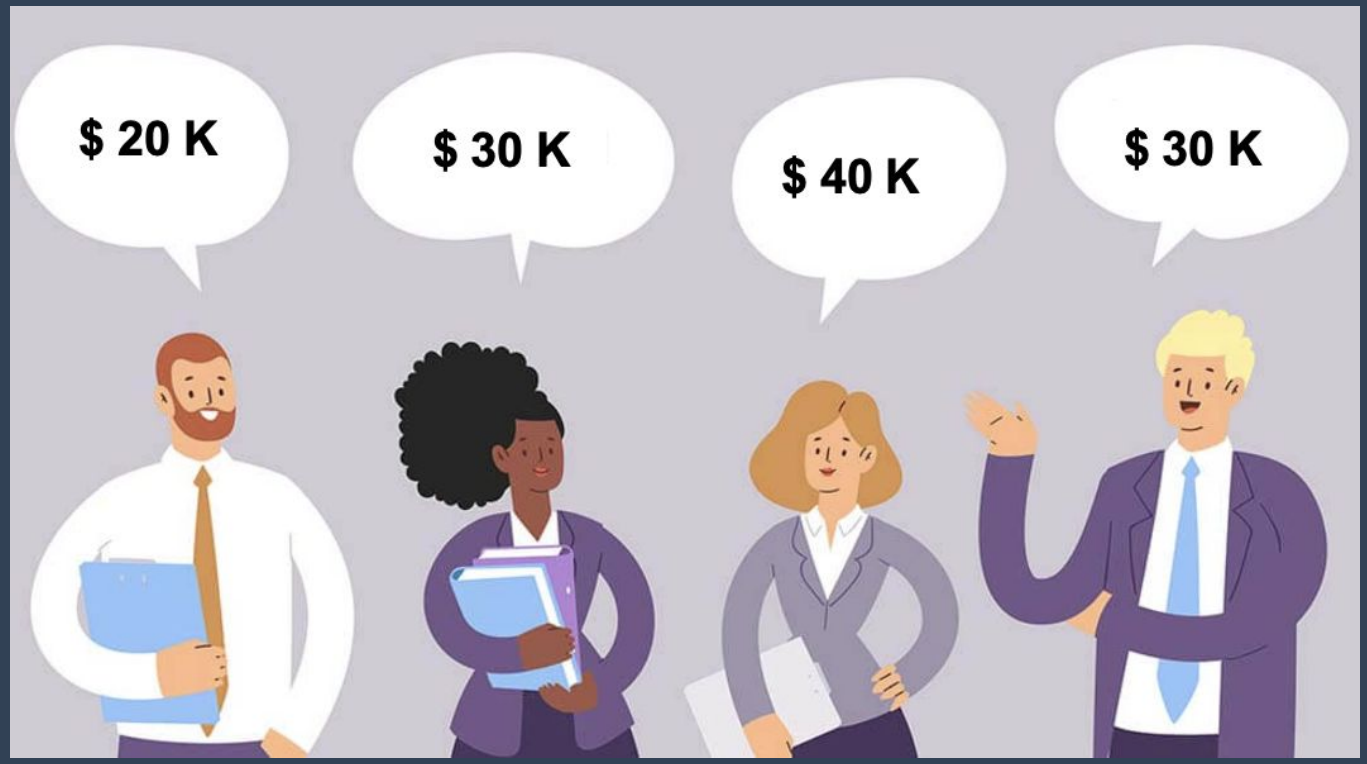
Median

- The median is the statistical value located at the center of a data set organized in ascending or descending order.
- For an odd-numbered of observations, the median is the observation that is located in the $(n+1)/2$ position.
- If the number of observations is even, the media is the mean value of the observations located in between $n/2$ & $(n+2)/2$ positions.



Median

Let’s find out ***Median Salary***.

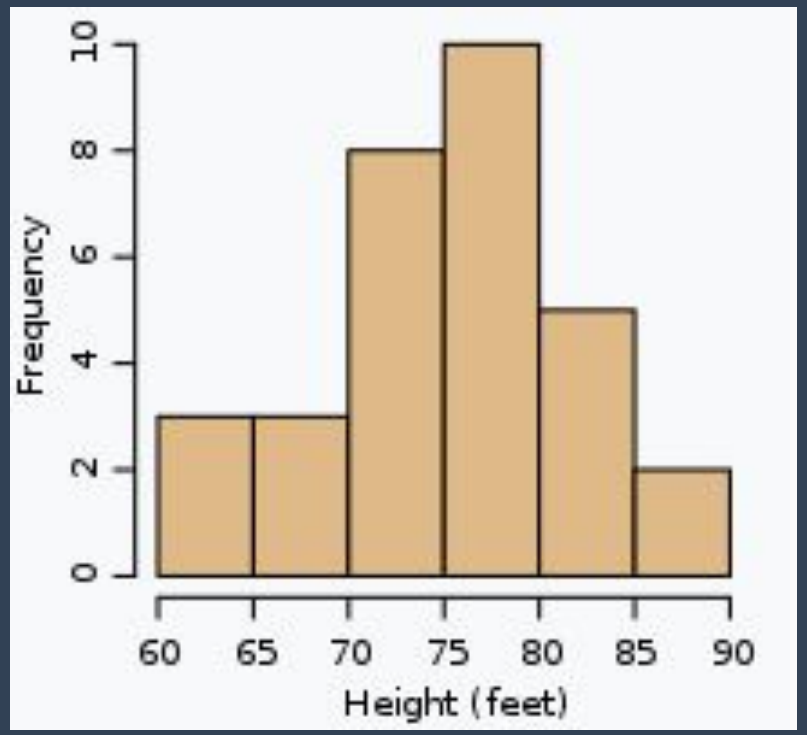


Mode

- The ***mode*** is the *value that appears most often (aka frequent) in a dataset*. Sometimes, a dataset has a mode; sometimes, it doesn't.

- If all the observations in a dataset are different and no value repeats more than others, then the dataset has no mode.

- A dataset with one mode is called unimodal. When there are two modes, it's called bimodal. If the distribution has three frequently occurring values, it's termed trimodal.



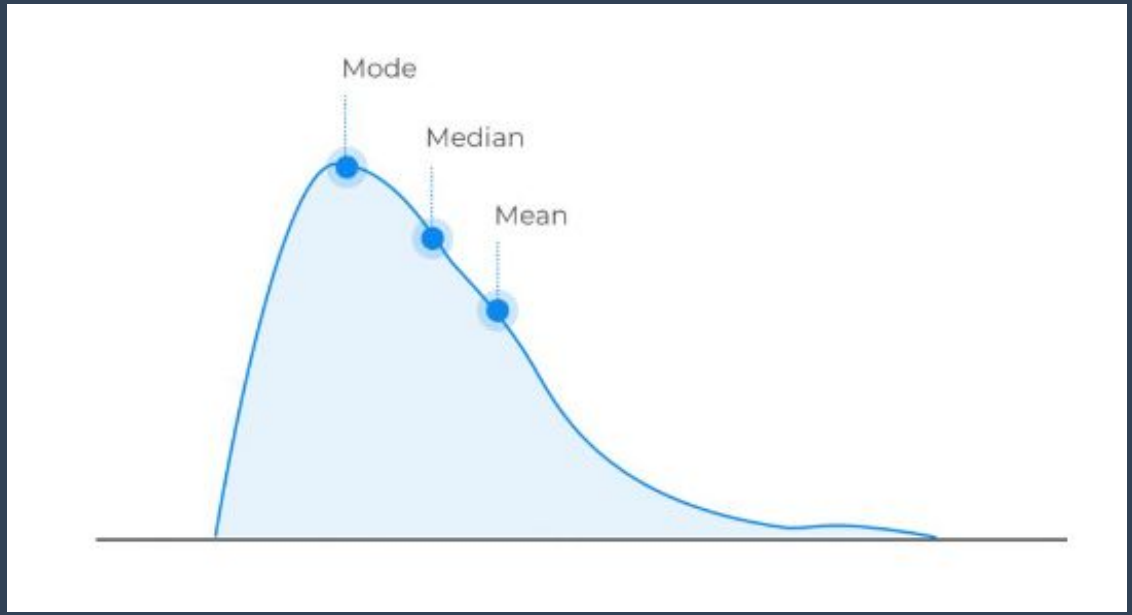
Mode

Let’s find out ***Mode of the Salary***.

Questions?

- Can mean = median = mode? When?

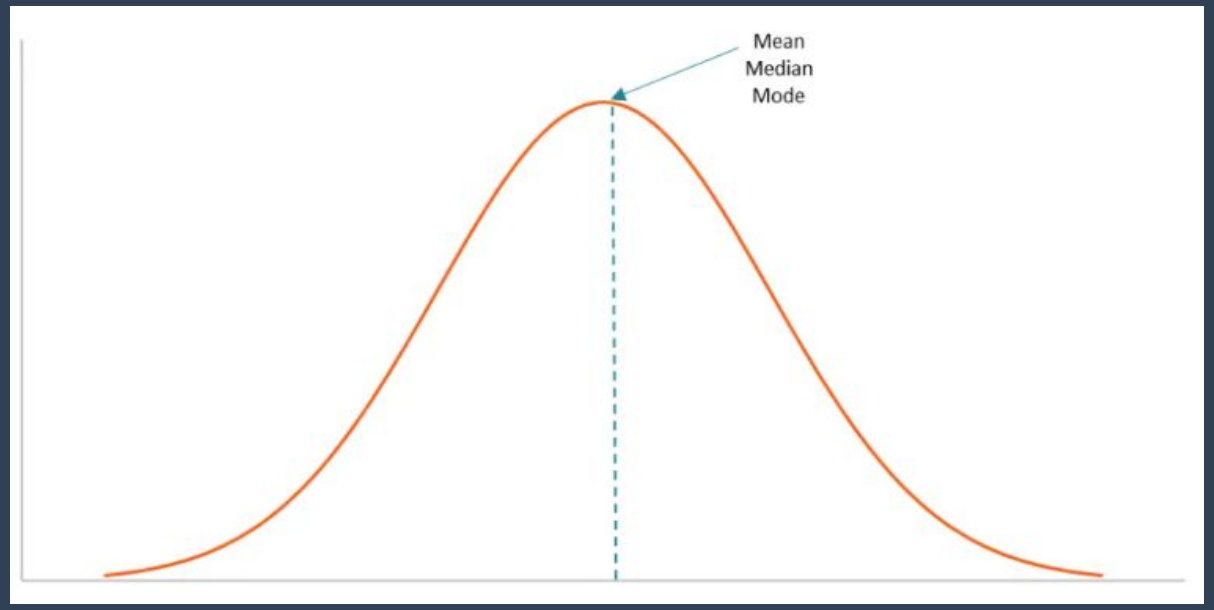
- When is median better than mean & when is mean better than median?



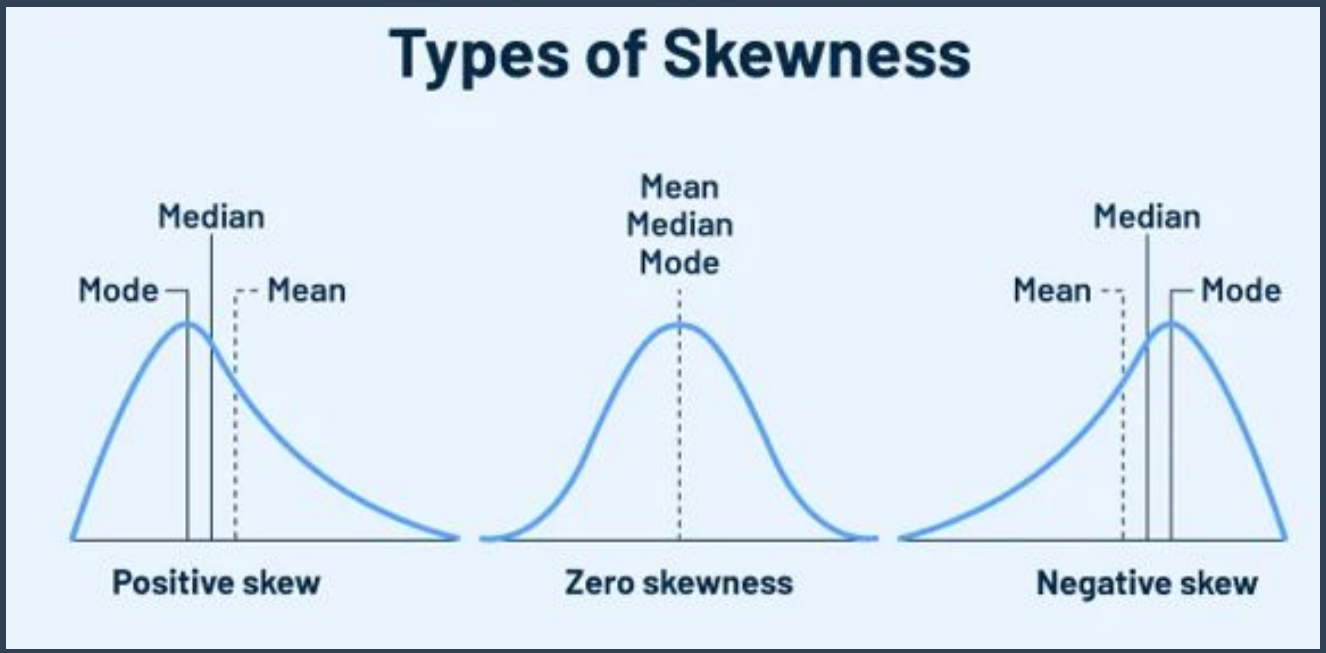
Questions?

- Can mean = median = mode? When?
  - Yes. Symmetric distribution with highest density in the center (e.g., Normal distribution)

- When is median better than mean & when is mean better than median?



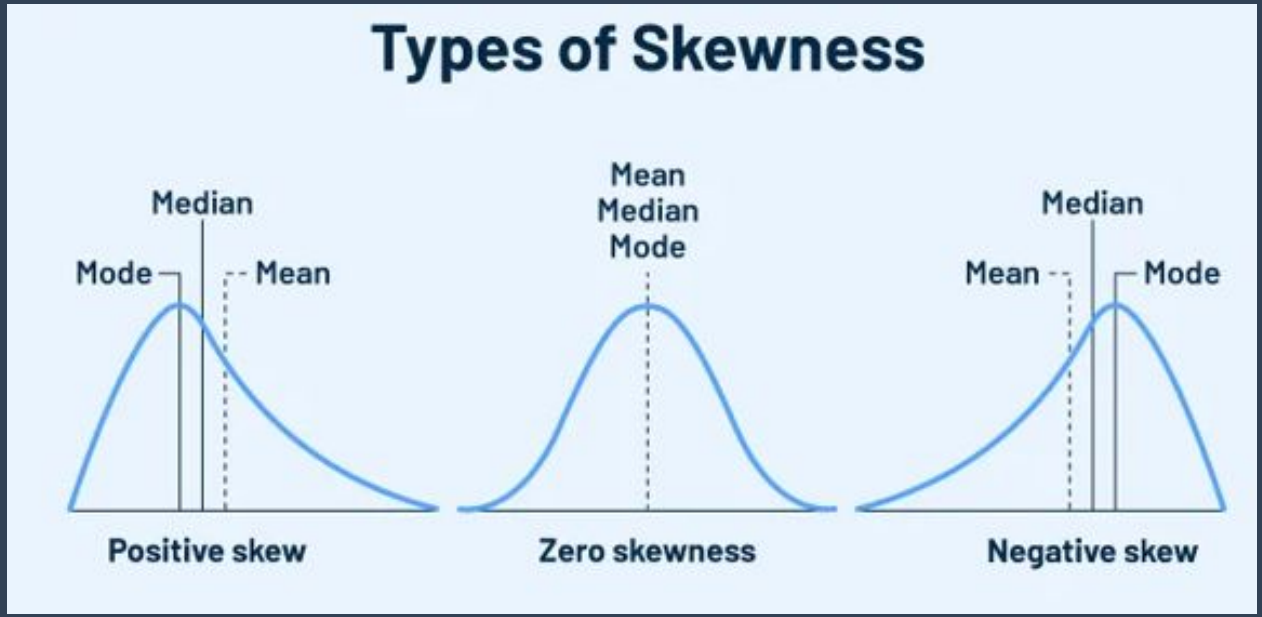
Skewness



- ***Skewness***: Measurement of asymmetry of the Distribution.
- ***Zero skewness***: Any symmetrical distribution - normal, uniform, and other distributions – has zero skews.
- ***Positive skewness***: A right-skewed distribution, or positive skew, has a long tail that inclines toward the right side.
- ***Negative skewness***: A left-skewed distribution, or negative skew, has a long tail that inclines towards the right side.

Skewness

- *Zero skewness*: Mean == Median
- *Positive skewness*: Mean > Median
- *Negative skewness*: Mean < Median



Questions?

- Can mean = median = mode? When?
  - *Yes. Symmetric distribution with highest density in the center (e.g., Normal distribution)*

- When is median better than mean? / When is mean better than median?
  - *Median is “robust to outlier”*
  - *Computational/analytical convenience*

Time to code